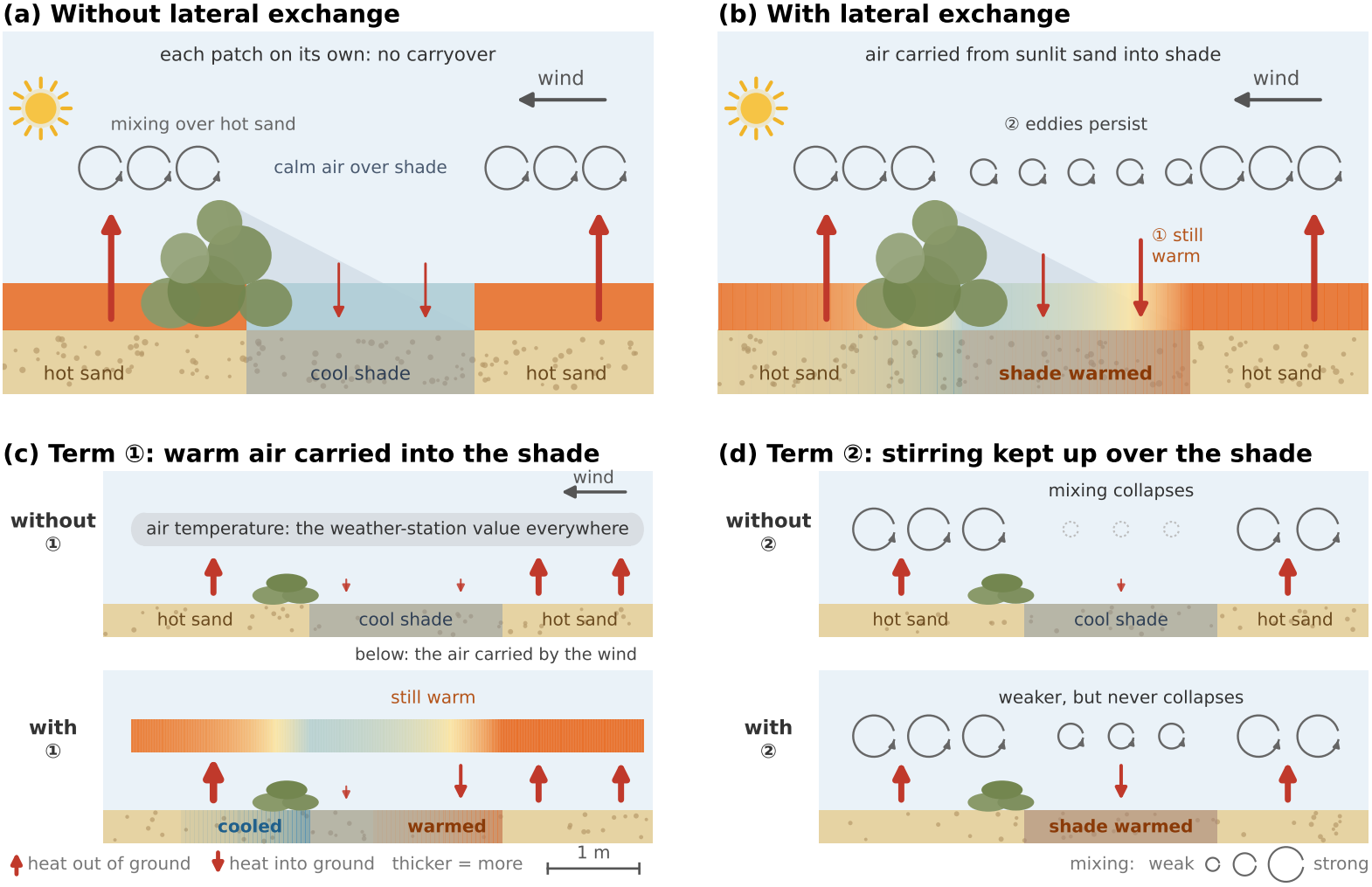

(a) Without lateral exchange
each patch on its own: no carryover
wind
mixing over hot sand
calm air over shade
hot sand
cool shade
hot sand
(b) With lateral exchange
air carried from sunlit sand into shade
wind
② eddies persist
① still warm
hot sand
shade warmed
hot sand
(c) Term ①: warm air carried into the shade
without ①
wind
air temperature: the weather-station value everywhere
hot sand
cool shade
hot sand
below: the air carried by the wind
with ①
still warm
cooled
warmed
heat out of ground
heat into ground
thicker = more
1 m
(d) Term ②: stirring kept up over the shade
without ②
mixing collapses
hot sand
cool shade
hot sand
with ②
weaker, but never collapses
shade warmed
mixing: weak strong